



CL-2001 Data Structures Lab # 11

Objectives:

- Binary Heap
- Max, Min Heap
- Heapify
- Heap Sort

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function about its functionality.
3. Use understandable name of variables.
4. Proper indentation of code is essential
5. Write a C++ statement(s) for each of the following task one after the other, in the same order.
6. Make a Microsoft Word file and paste all of your C++ code with all possible screenshots of every **task output in MS word and submit .cpp file with word file.**
7. Make separate .cpp files for all tasks and use this format **22F-1234_Task1.cpp.**
8. First think about statement problems and then write/draw your logic on copy.
9. After copy pencil work, code the problem statement on MS Studio C++ compiler.
10. At the end when you done your tasks, attached C++ created files in MS word file and make your submission on Google classroom. (Make sure your submission is completed).
11. Please submit your file in this format **22F-1234_L1.**
12. Do not submit your assignment **after the deadline.**
- 13. Do not copy code from any source otherwise you will be penalized with negative marks.**



Problem: 1 | Binary heap

- 1) **getMini():** It returns the root element of Min Heap.
- 2) **extractMin():** Removes the minimum element from MinHeap..
- 3) **insert():** Inserting a new key. We add a new key at the end of the tree. If new key is greater than its parent, then we don't need to do anything. Otherwise, we need to traverse up to fix the violated heap property.

Problem: 2 | Heap

A company which is responsible for the planning and construction of residential buildings in the city of Urbanville. The city has recently seen a spike in population, and there's a need for more residential buildings. The city has a specific policy about the construction of buildings; the building with the highest number of floors gets priority for construction.

A company has decided to use a heap data structure to efficiently handle this situation, where each building's priority is determined by the number of floors it has.

Ashok has the following operations he can perform:

addBuilding: This function will take the building's ID and the number of floors in that building as input and adds it to the construction queue.

constructBuilding: This function will construct the building with the highest priority (the building with the highest number of floors) and remove it from the queue.

getHighestPriorityBuilding: This function will return the ID of the building with the highest number of floors in the queue.

Problem: 3 | Convert BST to Min Heap

Given a binary search tree which is also a complete binary tree. The problem is to convert the given BST into a Min Heap.

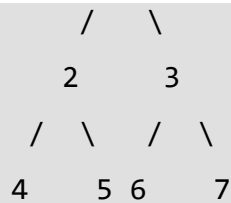
Examples:

Input:

```
      4
     / \
    2   6
   / \ / \
  1  3 5 7
```

Output:

```
      1
```



The given **BST** has been transformed into a **Min Heap**.

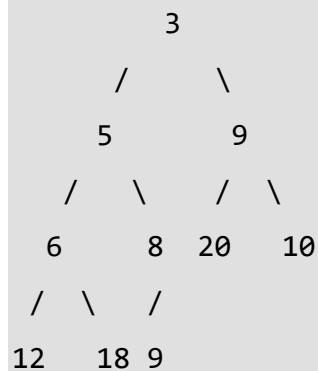
1. Create an array **arr[]** of size **n**, where **n** is the number of nodes in the given BST.
2. Perform the inorder traversal of the BST and copy the node values in the **arr[]** in sorted order.

While traversing the root during the inorder traversal, one by one copy the values from the array **arr[]** to the nodes.

Problem: 4 | Convert min Heap to max Heap

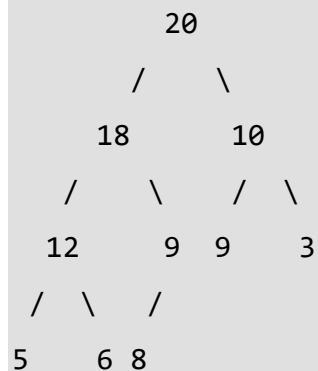
Given array representation of min Heap, convert it to max Heap. Example:

Input: `arr[] = [3 5 9 6 8 20 10 12 18 9]`



Output: `arr[] = [20 18 10 12 9 9 3 5 6 8]` OR

[any Max Heap formed from input elements]





The problem might look complex at first look. But our final goal is to only build the max heap. The idea is very simple – we simply build Max Heap without caring about the input. We start from bottom-most and rightmost internal node of min Heap and heapify all internal nodes in bottom up way to build the Max heap.

😊 Best of luck